

JAVA PROGRAMMING LAB (BCS154)

B.Tech CSE FIRST YEAR

SUBMITTED TO:

MR. ABHAY

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SN.	PROGRAM NAME	PAGE	DATE	SIGNATURE
1	Hello World, Variables and Arithmetic	1	26/10/2026	
2	Decisions and Loops with Numbers	3	26/10/2026	
3	Star and Number Patterns	5	26/10/2026	
4	Writing Functions	7	28/10/2026	
5	Arrays: Traversal and Counting	9	28/10/2026	
6	Arrays: Reverse and Duplicates	10	28/10/2026	
7	Counting Steps and Complexity	12	30/10/2026	
8	Matrices: Transpose and Row Sums	13	30/10/2026	
9	Matrices: Rows and Columns	15	30/10/2026	
10	Switch and ArrayList	17	02/11/2026	
11	Bubble Sort and Selection Sort	19	02/11/2026	
12	Insertion Sort	21	02/11/2026	
13	Strings: Methods and Palindromes	23	04/11/2026	
14	Characters: Counting and Positions	25	04/11/2026	
15	StringBuilder Methods	26	04/11/2026	

Assignment 1: Hello World, Variables and Arithmetic

Date: 26/10/2026

Aim: Print text, store values in variables and calculate with arithmetic operators.

Task 1: Print Hello World on one line and Welcome to Java on the next line.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 1.1
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World");
        System.out.println("Welcome to Java");
    }
}
```

Output:

```
Hello World
Welcome to Java
```

Task 2: Create variables of type int, double, char and String and print each with its label.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 1.2
public class Main {
    public static void main(String[] args) {
        int roll = 101;
        double percentage = 84.5;
        char section = 'B';
        String name = "Asha";
        System.out.println("Roll: " + roll);
        System.out.println("Percentage: " + percentage);
        System.out.println("Section: " + section);
        System.out.println("Name: " + name);
    }
}
```

Output:

```
Roll: 101
Percentage: 84.5
Section: B
Name: Asha
```

Task 3: Calculate sum, difference, product, quotient and remainder of a = 17 and b = 5.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 1.3
public class Main {
    public static void main(String[] args) {
        int a = 17, b = 5;
        System.out.println("Sum: " + (a + b));
        System.out.println("Difference: " + (a - b));
        System.out.println("Product: " + (a * b));
        System.out.println("Quotient: " + (a / b));
        System.out.println("Remainder: " + (a % b));
    }
}
```

Output:

```
Sum: 22
Difference: 12
Product: 85
Quotient: 3
Remainder: 2
```

Task 4: Use a final constant PI to print the area and perimeter of a circle of radius 7.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 1.4
public class Main {
    public static void main(String[] args) {
        final double PI = 3.14;
        int radius = 7;
        System.out.println("Area: " + (PI * radius * radius));
        System.out.println("Perimeter: " + (2 * PI * radius));
    }
}
```

Output:

```
Area: 153.86
Perimeter: 43.96
```

Assignment 2: Decisions and Loops with Numbers

Date: 26/10/2026

Aim: Use if, else if and loops to make decisions and repeat steps with whole numbers.

Task 1: Check whether n = 27 is even or odd using if-else.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 2.1
public class Main {
    public static void main(String[] args) {
        int n = 27;
        if (n % 2 == 0) {
            System.out.println(n + " is even");
        } else {
            System.out.println(n + " is odd");
        }
    }
}
```

Output:

```
27 is odd
```

Task 2: Print the grade for marks = 83 using an else-if ladder.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 2.2
public class Main {
    public static void main(String[] args) {
        int marks = 83;
        if (marks >= 90) {
            System.out.println("Grade: A");
        } else if (marks >= 75) {
            System.out.println("Grade: B");
        } else {
            System.out.println("Grade: C");
        }
    }
}
```

Output:

```
Grade: B
```

Task 3: Check whether the year 2024 is a leap year.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 2.3
public class Main {
    public static void main(String[] args) {
        int year = 2024;
        if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0) {
            System.out.println("Leap year");
        } else {
            System.out.println("Not a leap year");
        }
    }
}
```

Output:

```
Leap year
```

Task 4: Find the factorial of 5 using a for loop.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 2.4
public class Main {
```

```
    public static void main(String[] args) {
        int fact = 1;
        for (int i = 1; i <= 5; i++) {
            fact = fact * i;
        }
        System.out.println("Factorial of 5 is " + fact);
    }
}
```

Output:

```
Factorial of 5 is 120
```

Assignment 3: Star and Number Patterns

Date: 26/10/2026

Aim: Use a loop inside another loop to print stars and numbers in rows.

Task 1: Print a 4 x 4 square of stars.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 3.1
public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= 4; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

```
****
****
****
****
```

Task 2: Print a right triangle of stars with 4 rows.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 3.2
public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

```
*
**
***
****
```

Task 3: Print a number triangle with 4 rows.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 3.3
public class Main {
    public static void main(String[] args) {
        for (int i = 1; i <= 4; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print(j + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
1
1 2
```

```
1 2 3
1 2 3 4
```

Task 4: Print an upside-down triangle of stars with 4 rows.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 3.4
public class Main {
    public static void main(String[] args) {
        for (int i = 4; i >= 1; i--) {
            for (int j = 1; j <= i; j++) {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

Output:

```
****
***
**
*
```

Assignment 4: Writing Functions

Date: 28/10/2026

Aim: Write functions with parameters and return values, and call them from main.

Task 1: Write fact(n) and nCr(n, r), and print 5C2.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 4.1
public class Main {
    static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f = f * i;
        }
        return f;
    }

    static int nCr(int n, int r) {
        return fact(n) / (fact(r) * fact(n - r));
    }

    public static void main(String[] args) {
        System.out.println("5C2 = " + nCr(5, 2));
    }
}
```

Output:

```
5C2 = 10
```

Task 2: Write fact(n) and nPr(n, r), and print 5P2.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 4.2
public class Main {
    static int fact(int n) {
        int f = 1;
        for (int i = 1; i <= n; i++) {
            f = f * i;
        }
        return f;
    }

    static int nPr(int n, int r) {
        return fact(n) / fact(n - r);
    }

    public static void main(String[] args) {
        System.out.println("5P2 = " + nPr(5, 2));
    }
}
```

Output:

```
5P2 = 20
```

Task 3: Write maxOf(a, b) that returns the larger value.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 4.3
public class Main {
    static int maxOf(int a, int b) {
        if (a > b) {
            return a;
        }
        return b;
```

```
    }

    public static void main(String[] args) {
        System.out.println("Larger: " + maxOf(14, 9));
    }
}
```

Output:

```
Larger: 14
```

Assignment 5: Arrays: Traversal and Counting

Date: 28/10/2026

Aim: Create an array, visit every element with a loop, and add up or count the values.

Task 1: Print each index and element of an array using arr.length.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 5.1
public class Main {
    public static void main(String[] args) {
        int[] arr = {3, 4, 5};
        for (int i = 0; i < arr.length; i++) {
            System.out.println("Index " + i + ": " + arr[i]);
        }
    }
}
```

Output:

```
Index 0: 3
Index 1: 4
Index 2: 5
```

Task 2: Find the sum and the largest value of an array.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 5.2
public class Main {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 33, 21};
        int sum = 0, largest = arr[0];
        for (int i = 0; i < arr.length; i++) {
            sum += arr[i];
            if (arr[i] > largest) {
                largest = arr[i];
            }
        }
        System.out.println("Sum: " + sum);
        System.out.println("Largest: " + largest);
    }
}
```

Output:

```
Sum: 118
Largest: 45
```

Task 3: Count the values greater than 10.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 5.3
public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 12, 2, 14, 7};
        int count = 0;
        for (int i = 0; i < arr.length; i++) {
            if (arr[i] > 10) {
                count++;
            }
        }
        System.out.println("Count greater than 10: " + count);
    }
}
```

Output:

```
Count greater than 10: 2
```

Assignment 6: Arrays: Reverse and Duplicates

Date: 28/10/2026

Aim: Print an array in reverse order, find the smallest value, and check an array for repeated values.

Task 1: Print the array values in reverse order.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 6.1
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};
        for (int i = arr.length - 1; i >= 0; i--) {
            System.out.println(arr[i]);
        }
    }
}
```

Output:

```
50
40
30
20
10
```

Task 2: Find the smallest value and its index.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 6.2
public class Main {
    public static void main(String[] args) {
        int[] arr = {2, 5, 1, 8, 3, 4};
        int smallest = arr[0], index = 0;
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < smallest) {
                smallest = arr[i];
                index = i;
            }
        }
        System.out.println("Smallest: " + smallest + " at index " + index);
    }
}
```

Output:

```
Smallest: 1 at index 2
```

Task 3: Find a repeated value using nested loops.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 6.3
public class Main {
    public static void main(String[] args) {
        int[] arr = {4, 9, 2, 9, 7};
        boolean found = false;
        for (int i = 0; i < arr.length && !found; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    System.out.println("Repeated value: " + arr[i]);
                    found = true;
                    break;
                }
            }
        }
        if (!found) {
            System.out.println("No repeated value");
```

```
            }
        }
    }
```

Output:

```
Repeated value: 9
```

Assignment 7: Counting Steps and Complexity

Date: 30/10/2026

Aim: Count the steps a program takes, and write its time complexity as a comment.

Task 1: Count the steps of a single loop from 1 to 10.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 7.1
public class Main {
    public static void main(String[] args) {
        int steps = 0;
        // Time complexity: O(n)
        for (int i = 1; i <= 10; i++) {
            steps++;
        }
        System.out.println("Steps: " + steps);
    }
}
```

Output:

```
Steps: 10
```

Task 2: Count the steps of two nested loops from 1 to 10.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 7.2
public class Main {
    public static void main(String[] args) {
        int steps = 0;
        // Time complexity: O(n^2)
        for (int i = 1; i <= 10; i++) {
            for (int j = 1; j <= 10; j++) {
                steps++;
            }
        }
        System.out.println("Steps: " + steps);
    }
}
```

Output:

```
Steps: 100
```

Task 3: Count the halving steps of n = 1024.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 7.3
public class Main {
    public static void main(String[] args) {
        int n = 1024;
        int steps = 0;
        // Time complexity: O(log n)
        while (n > 1) {
            n = n / 2;
            steps++;
        }
        System.out.println("Halving steps: " + steps);
    }
}
```

Output:

```
Halving steps: 10
```

Assignment 8: Matrices: Transpose and Row Sums

Date: 30/10/2026

Aim: Create two-dimensional arrays, visit them row by row, and build a new matrix from one.

Task 1: Fill a 3 x 4 matrix with values 1 to 12 and print it.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 8.1
public class Main {
    public static void main(String[] args) {
        int[][] m = new int[3][4];
        int value = 1;
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                m[i][j] = value++;
            }
        }
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 4; j++) {
                System.out.print(m[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
1 2 3 4
5 6 7 8
9 10 11 12
```

Task 2: Create and print the transpose of a 3 x 3 matrix.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 8.2
public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        int[][] t = new int[3][3];
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                t[j][i] = m[i][j];
            }
        }
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                System.out.print(t[i][j] + " ");
            }
            System.out.println();
        }
    }
}
```

Output:

```
1 4 7
2 5 8
3 6 9
```

Task 3: Print the sum of each row of a 3 x 3 matrix.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 8.3
public class Main {
    public static void main(String[] args) {
```

```
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        for (int i = 0; i < 3; i++) {
            int sum = 0;
            for (int j = 0; j < 3; j++) {
                sum += m[i][j];
            }
            System.out.println("Row " + (i + 1) + " sum: " + sum);
        }
    }
}
```

Output:

```
Row 1 sum: 6
Row 2 sum: 15
Row 3 sum: 24
```

Assignment 9: Matrices: Rows and Columns

Date: 30/10/2026

Aim: Visit the rows and columns of a matrix with nested loops to print, add and count its values.

Task 1: Print the first row and the last row of a 3 x 3 matrix.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 9.1
public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        System.out.print("First row:");
        for (int j = 0; j < 3; j++) {
            System.out.print(" " + m[0][j]);
        }
        System.out.println();
        System.out.print("Last row:");
        for (int j = 0; j < 3; j++) {
            System.out.print(" " + m[2][j]);
        }
        System.out.println();
    }
}
```

Output:

```
First row: 1 2 3
Last row: 7 8 9
```

Task 2: Print the sum of each column of a 3 x 3 matrix.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 9.2
public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        for (int j = 0; j < 3; j++) {
            int sum = 0;
            for (int i = 0; i < 3; i++) {
                sum += m[i][j];
            }
            System.out.println("Column " + (j + 1) + " sum: " + sum);
        }
    }
}
```

Output:

```
Column 1 sum: 12
Column 2 sum: 15
Column 3 sum: 18
```

Task 3: Count the values greater than 5 in a 3 x 3 matrix.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 9.3
public class Main {
    public static void main(String[] args) {
        int[][] m = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
        int count = 0;
        for (int i = 0; i < 3; i++) {
            for (int j = 0; j < 3; j++) {
                if (m[i][j] > 5) {
                    count++;
                }
            }
```

```
        }
        System.out.println("Values greater than 5: " + count);
    }
}
```

Output:

```
Values greater than 5: 4
```

Assignment 10: Switch and ArrayList

Date: 02/11/2026

Aim: Choose a branch with a switch statement, and store and change a list of numbers in an ArrayList.

Task 1: Print the name of the day using a switch statement.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 10.1
public class Main {
    public static void main(String[] args) {
        int day = 3;
        switch (day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            default:
                System.out.println("Invalid day");
        }
    }
}
```

Output:

```
Wednesday
```

Task 2: Create an ArrayList and print the list, its size and the item at index 2.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 10.2
import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
        list.add(20);
        list.add(30);
        list.add(40);
        System.out.println(list);
        System.out.println("Size: " + list.size());
        System.out.println("Item at index 2: " + list.get(2));
    }
}
```

Output:

```
[10, 20, 30, 40]
Size: 4
Item at index 2: 30
```

Task 3: Change and delete items of an ArrayList using set and remove.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 10.3
import java.util.ArrayList;

public class Main {
    public static void main(String[] args) {
        ArrayList<Integer> list = new ArrayList<>();
        list.add(10);
```

```
        list.add(20);
        list.add(30);
        list.add(40);
        list.set(1, 25);
        list.remove(3);
        System.out.println(list);
        System.out.println("Size: " + list.size());
    }
}
```

Output:

```
[10, 25, 30]
Size: 3
```

Assignment 11: Bubble Sort and Selection Sort

Date: 02/11/2026

Aim: Sort small arrays with bubble sort and selection sort, and count the swaps.

Task 1: Sort the array 5, 1, 4, 2, 8 with bubble sort.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 11.1
public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                }
            }
        }
        System.out.print("Sorted: [");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i]);
            if (i < n - 1) {
                System.out.print(", ");
            }
        }
        System.out.println("]");
    }
}
```

Output:

```
Sorted: [1, 2, 4, 5, 8]
```

Task 2: Sort the array 64, 25, 12, 22, 11 with selection sort.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 11.2
public class Main {
    public static void main(String[] args) {
        int[] arr = {64, 25, 12, 22, 11};
        int n = arr.length;
        for (int i = 0; i < n - 1; i++) {
            int min = i;
            for (int j = i + 1; j < n; j++) {
                if (arr[j] < arr[min]) {
                    min = j;
                }
            }
            int temp = arr[min];
            arr[min] = arr[i];
            arr[i] = temp;
        }
        System.out.print("Sorted: [");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i]);
            if (i < n - 1) {
                System.out.print(", ");
            }
        }
        System.out.println("]");
    }
}
```

Output:

```
Sorted: [11, 12, 22, 25, 64]
```

Task 3: Count the swaps made by bubble sort on 5, 1, 4, 2, 8.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 11.3
public class Main {
    public static void main(String[] args) {
        int[] arr = {5, 1, 4, 2, 8};
        int n = arr.length;
        int swaps = 0;
        for (int i = 0; i < n - 1; i++) {
            for (int j = 0; j < n - 1 - i; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                    swaps++;
                }
            }
        }
        System.out.println("Swaps: " + swaps);
    }
}
```

Output:

```
Swaps: 4
```

Assignment 12: Insertion Sort

Date: 02/11/2026

Aim: Sort small arrays with insertion sort and count how many values are shifted.

Task 1: Sort the array 7, 3, 5, 1 with insertion sort.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 12.1
public class Main {
    public static void main(String[] args) {
        int[] arr = {7, 3, 5, 1};
        int n = arr.length;
        for (int i = 1; i < n; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.print("Sorted: [");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i]);
            if (i < n - 1) {
                System.out.print(", ");
            }
        }
        System.out.println("]");
    }
}
```

Output:

```
Sorted: [1, 3, 5, 7]
```

Task 2: Sort the array 14, 3, 9, 21, 6 with insertion sort.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 12.2
public class Main {
    public static void main(String[] args) {
        int[] arr = {14, 3, 9, 21, 6};
        int n = arr.length;
        for (int i = 1; i < n; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.print("Sorted: [");
        for (int i = 0; i < n; i++) {
            System.out.print(arr[i]);
            if (i < n - 1) {
                System.out.print(", ");
            }
        }
        System.out.println("]");
    }
}
```

Output:

```
Sorted: [3, 6, 9, 14, 21]
```

Task 3: Count the shifts made by insertion sort on 6, 5, 4, 3, 2, 1.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 12.3
public class Main {
    public static void main(String[] args) {
        int[] arr = {6, 5, 4, 3, 2, 1};
        int n = arr.length;
        int shifts = 0;
        for (int i = 1; i < n; i++) {
            int key = arr[i];
            int j = i - 1;
            while (j >= 0 && arr[j] > key) {
                arr[j + 1] = arr[j];
                shifts++;
                j--;
            }
            arr[j + 1] = key;
        }
        System.out.println("Shifts: " + shifts);
    }
}
```

Output:

```
Shifts: 15
```

Assignment 13: Strings: Methods and Palindromes

Date: 04/11/2026

Aim: Use common String methods, reverse a word, check a palindrome and print substrings.

Task 1: Print the length, position of g and upper case form of Programming.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 13.1
public class Main {
    public static void main(String[] args) {
        String word = "Programming";
        System.out.println("Length: " + word.length());
        System.out.println("Position of g: " + word.indexOf('g'));
        System.out.println("Upper case: " + word.toUpperCase());
    }
}
```

Output:

```
Length: 11
Position of g: 3
Upper case: PROGRAMMING
```

Task 2: Reverse the word Java using charAt and +.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 13.2
public class Main {
    public static void main(String[] args) {
        String word = "Java";
        String rev = "";
        for (int i = word.length() - 1; i >= 0; i--) {
            rev = rev + word.charAt(i);
        }
        System.out.println("Reversed: " + rev);
    }
}
```

Output:

```
Reversed: avaJ
```

Task 3: Check whether racecar is a palindrome.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 13.3
public class Main {
    public static void main(String[] args) {
        String word = "racecar";
        boolean palindrome = true;
        for (int i = 0; i < word.length() / 2; i++) {
            if (word.charAt(i) != word.charAt(word.length() - 1 - i)) {
                palindrome = false;
                break;
            }
        }
        if (palindrome) {
            System.out.println(word + " is a palindrome");
        } else {
            System.out.println(word + " is not a palindrome");
        }
    }
}
```

Output:

```
racecar is a palindrome
```

Task 4: Print every substring of abc using substring().

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 13.4
public class Main {
    public static void main(String[] args) {
        String word = "abc";
        for (int i = 0; i < word.length(); i++) {
            for (int j = i + 1; j <= word.length(); j++) {
                System.out.println(word.substring(i, j));
            }
        }
    }
}
```

Output:

```
a
ab
abc
b
bc
c
```

Assignment 14: Characters: Counting and Positions

Date: 04/11/2026

Aim: Use charAt to count vowels and a letter, and print each letter with its position.

Task 1: Count the vowels in programming.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 14.1
public class Main {
    public static void main(String[] args) {
        String s = "programming";
        int count = 0;
        for (int i = 0; i < s.length(); i++) {
            char c = s.charAt(i);
            if (c == 'a' || c == 'e' || c == 'i' || c == 'o' || c == 'u') {
                count++;
            }
        }
        System.out.println("Vowels: " + count);
    }
}
```

Output:

```
Vowels: 3
```

Task 2: Count how many times g appears in programming.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 14.2
public class Main {
    public static void main(String[] args) {
        String s = "programming";
        int count = 0;
        for (int i = 0; i < s.length(); i++) {
            if (s.charAt(i) == 'g') {
                count++;
            }
        }
        System.out.println("Count of g: " + count);
    }
}
```

Output:

```
Count of g: 2
```

Task 3: Print each letter of hello with its position.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 14.3
public class Main {
    public static void main(String[] args) {
        String s = "hello";
        for (int i = 0; i < s.length(); i++) {
            System.out.println(i + ": " + s.charAt(i));
        }
    }
}
```

Output:

```
0: h
1: e
2: l
3: l
4: o
```

Assignment 15: StringBuilder Methods

Date: 04/11/2026

Aim: Build and change text with StringBuilder, using append, insert and deleteCharAt.

Task 1: Append World to Hello and print the text and its length.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 15.1
public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Hello");
        sb.append(" World");
        System.out.println(sb);
        System.out.println("Length: " + sb.length());
    }
}
```

Output:

```
Hello World
Length: 11
```

Task 2: Append 1 to 5 with commas and remove the last comma.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 15.2
public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder();
        for (int i = 1; i <= 5; i++) {
            sb.append(i).append(",");
        }
        sb.deleteCharAt(sb.length() - 1);
        System.out.println(sb);
    }
}
```

Output:

```
1,2,3,4,5
```

Task 3: Insert Hello at the start of Java using insert.

Program:

```
// Name: Aakriti Sharma | Roll No: BCS2026277 | Assignment 15.3
public class Main {
    public static void main(String[] args) {
        StringBuilder sb = new StringBuilder("Java");
        sb.insert(0, "Hello ");
        System.out.println(sb);
    }
}
```

Output:

```
Hello Java
```